1 What does the prefix *tera* represent?

$$\frac{\delta Q}{Q} = \left[\frac{1}{2}\left(\frac{\delta x}{x}\right) + \frac{1}{2}\left(\frac{\delta y}{y}\right) + \left(\frac{\delta z}{z}\right)\right]$$

A 10^{15}
B 10^{12}
C 10^{9}
D 10^{6}

2 A quantity Q is given by $Q = k\sqrt{\frac{xyz^2}{z^3}}$, where k is a dimensionless constant.

The percentage uncertainties in x, y, and z are $\pm 0.6\%$, $\pm 0.3\%$ and $\pm 0.4\%$ respectively.

What is the percentage uncertainty in Q?

~~$\frac{\delta Q}{Q} \times 100 = \pm 0.6$~~

A $\pm 0.85\ \%$
B $\pm 1.2\ \%$
C $\pm 1.3\ \%$
D $\pm 2.4\ \%$

$0.3 + \frac{0.3}{2},\ (0.4)\frac{\times 2}{2}$

3 A particle was projected upwards with a velocity, U, at an angle, θ, to the vertical.

What was the magnitude of the velocity of the particle when it was at a height, h, above the ground?

A $U\sin\theta$
B $\left(U^2\cos^2\theta - 2gh\right)^{\frac{1}{2}}$
C $\left(U^2 - 2gh\right)^{\frac{1}{2}}$
D $\left(U^2\sin^2\theta - 2gh\right)^{\frac{1}{2}}$

$v^2 = u^2 + 2as$
$v^2 = (u\cos\theta)^2 + 2(-g)h$
$v = (U^2\cos^2\theta - 2gh)^{\frac{1}{2}}$

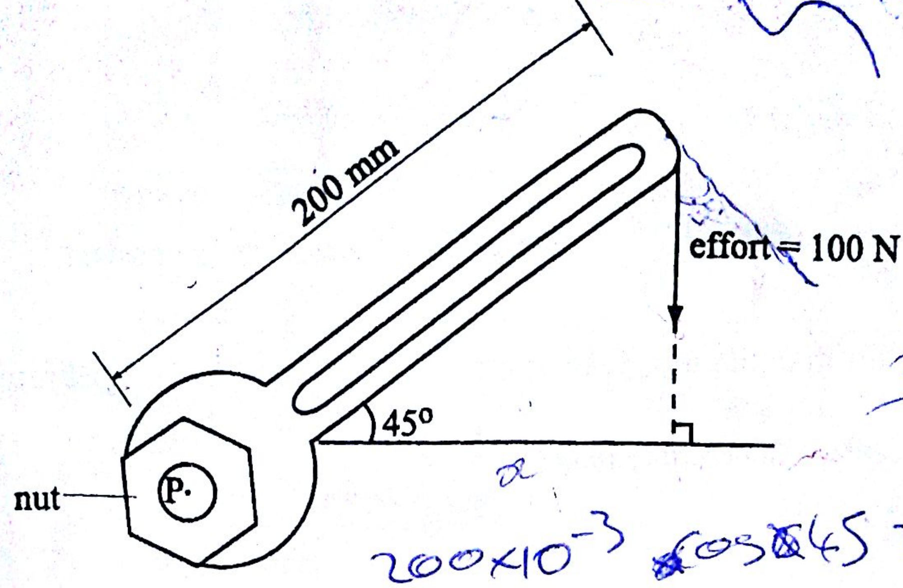
4 A stunt rider leaves a ramp at an angle of $30°$ above the horizontal and lands at the same height, h, after travelling a horizontal distance of 36 m.

What was the take off speed?

$s = ut$
$36 = v\cos\theta\, t$
$\frac{36}{v\cos\theta}$
$0 = v\sin\theta\, t - \frac{1}{2}gt^2$
$0 = \left(\frac{36}{v\cos\theta}\right)(v\sin\theta) - \frac{1}{2}g\left(\frac{36}{v\cos\theta}\right)^2$

A $10\ \mathrm{ms^{-1}}$
B $12\ \mathrm{ms^{-1}}$
C $14\ \mathrm{ms^{-1}}$
D $20\ \mathrm{ms^{-1}}$

5 The diagram shows a spanner used to remove a nut.



$200\times10^{-3}\cos 45\times100$

The moment of the effort about P is

A 20 000 Nm.

B 14 000 Nm.

C 20 Nm.

D 14 Nm.

In a head-on collision between objects A and B, moving in opposite directions, the

A momentum of A is opposite that of B.

B total momentum of the system stays constant.

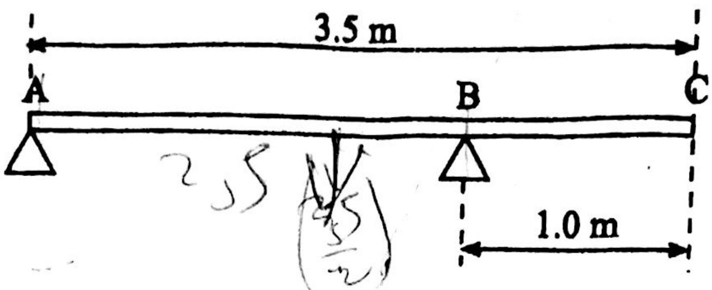
C initial and final momentum of A is the same.

D initial and final momentum of B is the same.

7 A 40 kg uniform beam of length 3.5 m has two supports A and B as shown.

$\frac{\lambda}{\lambda}$

54%



The normal reaction at A is approximately equal to $A(3.5) = \left(\frac{3.5}{2} - 1\right)40g$

A 100 N
B 120 N
C 160 N
D 280 N

8 A 400 000 kg train has a top speed of 50 ms^{-1} when its engine power is 2 MW.

What is the kinetic energy of the train at its top speed? $\frac{1}{2}(4\times10^{6})(50)^{2}$

A 20 MJ
B 40 MJ
C 100 MJ
D 500 MJ

400 000 kg

9 What is the condition for constructive interference?

A The path difference must be a whole number multiple of the wavelength.
B The distance between the screen and the sources of waves must be large.
C The distance between sources of waves must be large.
D The path difference must be an odd number multiple of half wavelengths.

10 For which system does the force between two objects vary inversely as the square of the distance between their centres?

A two long straight parallel conductors carrying steady currents
B two equal masses joined by a rubber band under tension
C a satellite orbiting about the Earth
D two flat circular coils carrying a steady current

11 A satellite moving with constant speed, v, in a circular orbit of radius, r, completes one revolution in time, t.

What is its centripetal acceleration?

$\frac{v^2}{r}$ $r\left(\frac{2\pi}{T}\right)^2$ $\frac{4\pi^2 r}{t^2}$

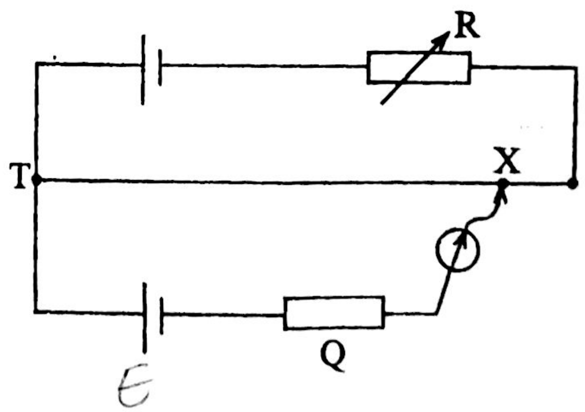
A $\dfrac{2\pi r}{t^2}$

B $\dfrac{4\pi^2 r^2}{t}$

C $\dfrac{2\pi r}{t}$

D $\dfrac{4\pi^2 r}{t^2}$

12 The diagram shows a potentiometer circuit with a balance point X.



A balance point closer to T can be obtained by

A replacing the test cell by one with a greater e.m.f.
B increasing the thickness of the potentiometer wire.
C reducing the thickness of the potentiometer wire.
D replacing the test cell by one with a lower e.m.f.

13 X-rays are produced by energy changes by the

A electrons in the outer shell.
B electrons in the inner shell.
C electrons and neutrones.
D electrons and protons.

14 Why should an ammeter be well damped?

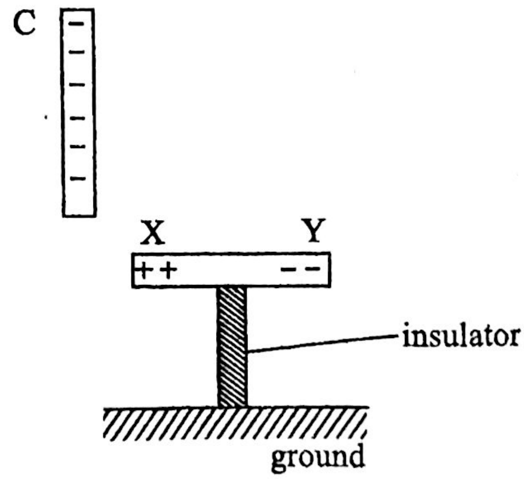
A for it to be very accurate
B for it to measure small currents
C for readings to be taken quickly
D for it to be very robust

15 One farad is the same as

A one coulomb per volt.
B one ohm per meter.
C one volt per coulomb.
D one joule per coulomb.

$C = \frac{Q}{V}$

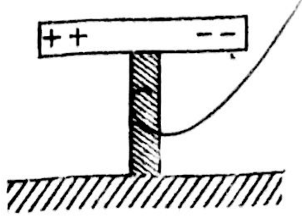
16 A negatively charged polythene rod, C, is brought near an insulated conductor XY as shown in the diagram.



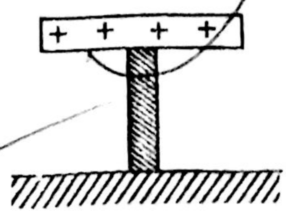
The conductor was touched at Y by a bare hand with C in the same position.

Which diagram shows the distribution of charge in XY when C is removed?

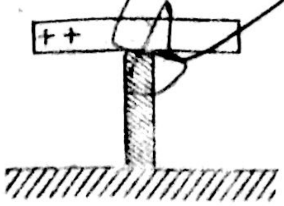
A



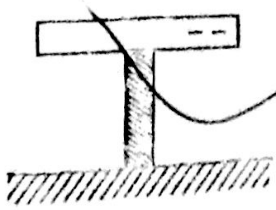
B



C



D



17 Which is **not** an application of X-rays?

A imaging of internal body structures
B treatment of malignancy
C sterilisation of surgical equipment
D geological dating of specimens

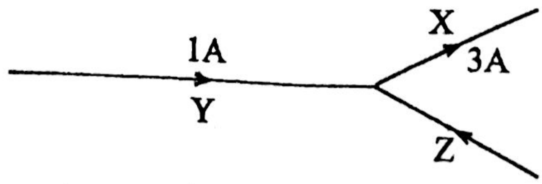
18 An electron is deflected in a uniform magnetic field of strength 6×10^{-3} T into a circular path of radius 40 mm.

What is the velocity of the electron?

A 2.3×10^{4} m/s
B 4.2×10^{7} m/s
C 2.4×10^{8} m/s
D 4.2×10^{10} m/s

$B = 6 \times 10^{-3}$
$r = 40 \times 10^{-3}$
$\frac{mv^2}{r} = Bqv$
$v^2 = \sqrt{\frac{Bqr}{m}}$
$v =$

19 The diagram shows a junction of three current carrying wires X, Y, and Z.



What is the charge passing a point in wire Z in 2 seconds?

A 1 C
B 2 C
C 4 C
D 8 C

$3 + Z = 1$
$Z = 2$
2×2
$4C$

20 When a beam of monochromatic light of a wavelength, λ, is incident normally on a diffraction grating with spacing, d, the second order maximum is at an angle, θ, to the normal.

$\sin\theta$ is given by

d

$d\sin\theta = n\lambda$

$\sin\theta = \left(\frac{2\lambda}{d}\right)$

A $\frac{2d}{\lambda}$.

B $\frac{d}{2\lambda}$.

C $\frac{2\lambda}{d}$.

D $\frac{\lambda}{d}$.

21 The effects associated with electromagnetic induction oppose the change that causes them.

This is because of the conservation of

A energy.
B momentum.
C mass.
D charge.

22 A current balance may be used to define the unit of electric current.

On what effect of current does the operation of the balance depend?

A chemical
B heating
C lighting
D magnetic

23 Which statement about an ideal operational amplifier is not correct?

A It has an infinite slew rate.
B It has an infinite bandwidth.
C Both alternating current and direct current can be amplified.
D It has an infinitely low input impedance.

24 A transformer used to step up 240 V to 600 V has a primary coil of 100 turns. What is th number of turns on the secondary coil?

A 40
B 150
C 250
D 1 440

$\frac{600}{240} \times 100 = N$

25 In a uniform electric field, all

A charged particles experience the same force.
B charged particles move with the same velocity.
C electric field lines are directed towards positive charge.
D electric field lines are parallel to each other.

26 A failure due to sustained stress below that required for immediate failure combined with elevated temperatures is called

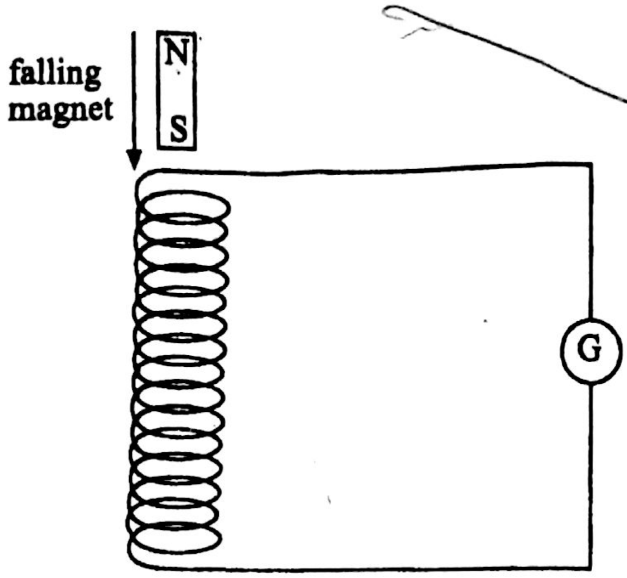
A creep.
B fatigue.
C breaking stress.
D cyclic stress.

27 A pupil wants to design a circuit in which a burglar alarm sounds when it is dark and a door is opened.

Which single logic gate does she use?

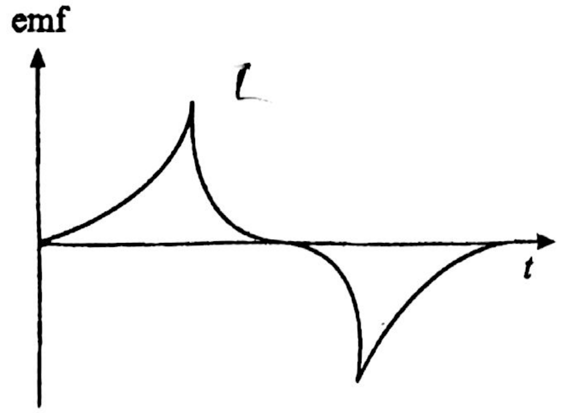
A OR
B AND
C NAND
D NOR

28 The diagram is used to illustrate electromagnetic induction.

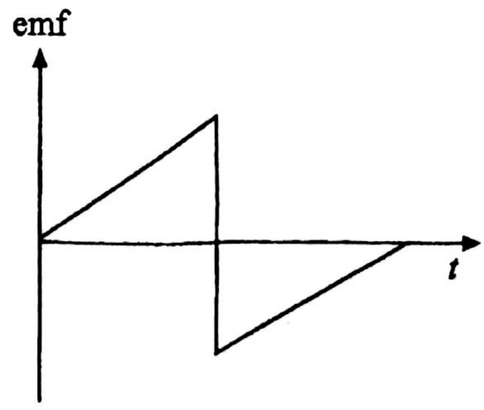


Which graph shows the variation of the induced e.m.f with time, *t*?

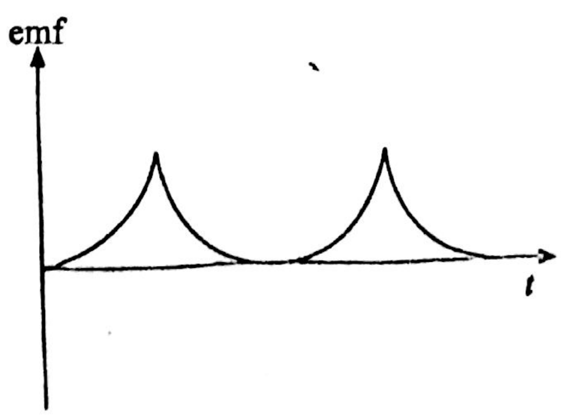
A



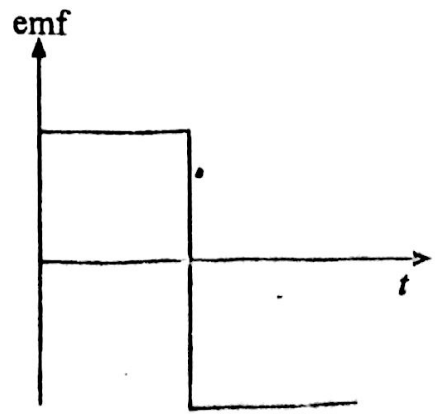
B



C



D



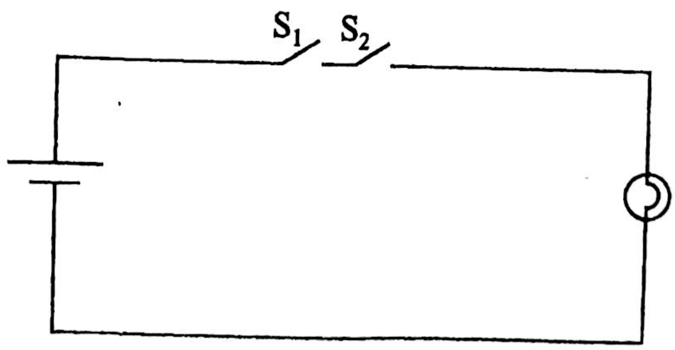
29 The stuff inside a hot pie is hotter than the crust because

A the stuff has a higher specific heat capacity than the crust.
B the crust has a higher specific heat capacity than the stuff.
C the stuff has a higher heat capacity than the crust.
D the crust has a higher heat capacity than the stuff.

30 Which statement is true about the thermodynamic temperature scale?

A It uses one fixed point, the triple point of water.
B It uses two fixed points, the ice point and the steam point.
C It uses two fixed points, the absolute zero and the triple point of water.
D It uses one fixed point, the absolute zero temperature.

31 The diagram shows switches S_1 and S_2, a bulb and a cell connected in series.



Which logic gate performs the same function as the circuit?

A OR
B NOR
C AND
D NAND